Cross-Basin Generalization for Tropical Cyclone Forecasting using TropiCycloneNet

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Problem Context

Tropical cyclone (TC) forecasting in data-scarce ocean basins is limited by small training sets. The South Pacific (SP) contains 5× fewer storms than the Western Pacific (WP) in the TropiCycloneNet dataset (30 vs. 131 storms), making it impractical to train reliable models on SP alone [1]. Transfer learning from data-rich to data-scarce basins is a natural solution, yet a fundamental physical barrier exists: the Coriolis force reverses across hemispheres, causing WP storms to track NW/W while SP storms track SE/W/SW.

Despite this directional shift, the underlying atmospheric physics governing TC intensity, sea-surface temperature (SST) and vertical wind shear, is shared across basins [3]. A physics-aware model that disentangles hemisphere-dependent kinematics from universal thermodynamics should therefore transfer effectively with minimal adaptation. We encode the Coriolis reversal through a phase shift on directional labels, following evidence that WP-only models fail on Southern Hemisphere basins without hemisphere-aware preprocessing [1].

Proposed Approach

We evaluate **five architectures** on zero-shot WP→SP transfer, then fine-tune the best performers:

Model	Transfer hypothesis
U-Net	Spatial baseline; strong 3D field reconstruction
ResNet	Deep residual features; proven image classifier
FNO	Mesh-invariant; spectral basin-agnostic maps
PI-GAN	Adversarial + vorticity/PV conservation
Transformer	Cross-attention fusion of 3 modalities

Three-stage pipeline:

1. **Zero-shot**: Train on WP, apply Coriolis phase shift, evaluate on SP.
2. **Fine-tune**: Adapt top models on 10/25/50% of SP data; measure accuracy vs. compute.
3. **Extend**: Test generalization to NI and SI basins.

Dataset

TropiCycloneNet [1] covers 6 ocean basins (1950–2023) with three modalities at 6-hourly resolution: **Data1D** (cyclone-center time series from IBTrACS [5]), **Data3D** (81×81 gridded SST, u/v wind, geopotential at 0.25°), and **Env-Data** (structured environmental features).

	WP	**SP**	**Ratio**
Storms	131	30	4.4×
Timesteps	4,562	922	5.0×
Top directions	NW/W	SE/W	Mirrored
Mean shear	8.2 m/s	7.4 m/s	Similar

Wind shear and SST distributions overlap substantially between WP and SP (Fig. 4), confirming that universal intensity predictors [3] should transfer. Direction distributions are hemisphere-mirrored (Fig. 3), validating our Coriolis phase-shift preprocessing.

References [1] Huang et al., "Benchmark dataset and deep learning method for global tropical cyclone forecasting," *Nat. Commun.*, 2025. [2] Qu et al., "Accurate tropical cyclone intensity forecasts using a non-iterative spatiotemporal transformer model," *npj Clim. Atmos. Sci.*, 2025. [3] DeMaria & Kaplan, "A Statistical Hurricane Intensity Prediction Scheme (SHIPS)," *Wea. Forecasting*, 1994. [4] Chen et al., "FourCastNet: Adaptive Fourier Neural Operators for weather forecasting," *arXiv:2202.11214*, 2022. [5] Knapp et al., "IBTrACS," *Bull. Amer. Meteor. Soc.*, 2010.

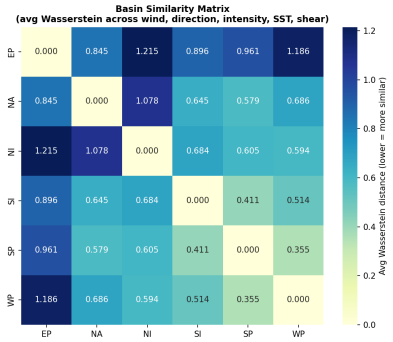

Fig. 1: Storm counts per basin. SP has 5× fewer storms than WP, motivating transfer learning.

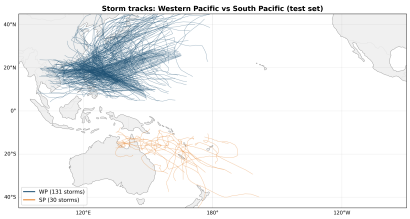

Fig. 2: WP vs SP storm tracks. Geographic separation and hemisphere flip are clearly visible.

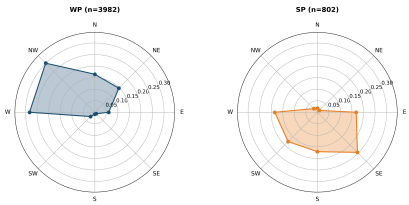

Fig. 3: Direction polar roses showing hemisphere-mirrored movement patterns (WP: NW/W; SP: SE/W).

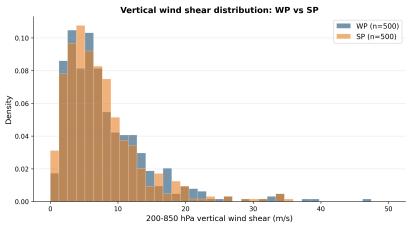

Fig. 4: SST and shear distributions overlap across basins—physics transfers despite directional shift.